James Akl

Robotics Researcher

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COURSEWORK WPI Transcript | LAU Transcript

Select coursework is tabulated below. Transcripts are linked above to view all courses. Codes are listed to verify courses on transcript.

Abbreviations GR = *Graduate* UG = *Undergraduate* WPI = *Worcester Polytechnic Institute* LAU = *Lebanese American University*

TOPIC GROUP	TOPIC	LEVEL	INSTITUTION	GRADE	COURSE CODES
Robotic Systems	Foundations of Robotics	GR	WPI	A	RBE 500
	Robot Dynamics	GR	WPI	A	RBE 501
	Robot Control	GR	WPI	A	RBE 502
	Biomechanics & Robotics	GR	WPI	A	RBE 520
	Robot & AI Ethics	GR	WPI	–	WR 593
Pure Mathematics	Convex Analysis & Optimization	GR	LAU	A	ACM 720
	Real Analysis	UG	LAU	A	MTH 401
	Discrete Structures	UG	LAU	A	MTH 207
Applied Mathematics	Computational Inverse Problems	GR	WPI	A	MA 590
	Applied Statistics	GR	WPI	A	MA 511
	Engineering Mathematics	GR	WPI	A	MA 501
	Numerical Linear Algebra	GR	LAU	A	ACM 713
	Numerical Methods	UG	LAU	A	GNE 333
	Probability & Statistics	UG	LAU	A	GNE 331
	Multivariate Calculus (4 courses)	UG	LAU	A	MTH 101, 102, 201, 206
	Differential Equations	UG	LAU	A	MTH 304
	Finite Element Methods	UG	LAU	A	MEE 421
	Optimization	UG	LAU	A	INE 420
Software Systems	Advanced Cryptography	GR	WPI	A	ECE 673
	Python Programming	GR	WPI	A	CS 5007
	Object-Oriented Programming	UG	LAU	A	COE 212
	MATLAB Programming	UG	LAU	A	MEE 212
Management Science	Entrepreneurship & Innovation	UG	LAU	A	ETR 500
	Engineering Economy	UG	LAU	A	INE 320
	Microeconomics	UG	LAU	A	ECO 201
	Project Management	UG	LAU	B	INE 428
System Design	Control Systems	UG	LAU	A	MEE 445, 446
	Instrumentation & Measurements	UG	LAU	B	MEE 391
	Electrical Circuits	UG	LAU	B	ELE 305
	Computer Aided Design	UG	LAU	A	MEE 351
	Engineering Graphics	UG	LAU	A	MEE 211
Mechanical Systems	Dynamics & Statics (3 courses)	UG	LAU	A	PHY 111, CIE 200, MEE 241
	Mechanical Vibrations	UG	LAU	A	MEE 442, 443
	Kinematics of Linkages	UG	LAU	B	MEE 341
	Mechanical Design	UG	LAU	B	MEE 422
Thermal Systems	Fluid Mechanics	UG	LAU	B	MEE 311, 312
	Thermodynamics (2 courses)	UG	LAU	B	MEE 301, 302
	Heat Transfer	UG	LAU	B	MEE 403, 404
	Internal Combustion Engines	UG	LAU	A	MEE 407
	Refrigeration & Air Conditioning	UG	LAU	B	MEE 515, 516
	Thermal System Design	UG	LAU	B	MEE 414
Materials Science	General Chemistry (2 courses)	UG	LAU	A	CHM 101, 200
	Material Properties	UG	LAU	A	MEE 321
	Strength of Materials	UG	LAU	B	MEE 320
	Manufacturing Processes	UG	LAU	B	MEE 332, 333